PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf)** and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Clinical practice guidelines for the management of adult patients with neurogenic lower urinary tract dysfunction: a systematic review protocol
AUTHORS	Zhao, Chun; Hu, YingJie; Wang, Xiaojiao; Hao, Fengming; Wang, Ying; Chen, Ling; Cai, Wen Zhi

VERSION 1 – REVIEW

REVIEWER	Medina-Polo, Jose Univ Hosp 12 Octubre, Urology
REVIEW RETURNED	06-Jul-2022

GENERAL COMMENTS	None

REVIEWER	Walter, Matthias Spinal Cord Injury Center & Research, University of Zurich, Neuro-Urology
REVIEW RETURNED	19-Jul-2022

GENERAL COMMENTS	The authors present a protocol to conduct a systematic review on clinical practice guidelines for the neurogenic lower urinary tract dysfunction (NLUTD). Their aim is clear, their methodology is sound, and their strengths and limitations are well balanced. Given the current paucity on that topic, their future systematic review will hopefully provide well-needed evidence to improve patient care. I have only minor comments. Minor comments: Page 2, lines 23-25: I suggest to revise the sentences to: Neurogenic lower urinary tract dysfunction (NLUTD) not only threatens the health of affected patients long-term but also has a significantly negative impact on the patients' quality of life (QoL). Page 2, line 26: The authors use 'extant' several times throughout the manuscript. This reviewer feels that the author might not use the right word to express their intentions. Thus, the authors should either please explain what they are trying to indicate by using "extant" or use a different term instead. - Page 3, line 54 - Page 5, line 106 - Page 6, line 133 - Page 6, line 137

	Page 3, line 44: With respect to “The findings of this study will be published in peer-reviewed journals”, this reviewer feels that the authors meant to use journal, i.e. published in a peer-reviewed journal, rather than in multiple journals Page 3, line 46: With respect to the PROSPERO registration, there is a significant difference between the authors highlighted on PROSPERO compared to the current protocol submission. This should be clarified by the authors. Page 3, line 51: With respect to “focused on a single population with NLUTD after spinal cord injury”, rather use focused on a single population with NLUTD, i.e. after spinal cord injury. Page 4, lines 70-72: I suggest to revise the sentences, since “neurogenic bladder” is outdated and should not be used anymore, to: Neurogenic Lower Urinary Tract Dysfunction (NLUTD) is defined by the International Continence Society (ICS) as a “lower urinary tract dysfunction due to disturbance of the neurological control mechanism”. Page 4, lines 73-77: Please revise this section “Studies show that 70%-84% of patients with spinal cord injury (SCI),2 53% of stroke patients,3 38%-71% of patients with Parkinson's disease, 32%-96% of patients with multiple sclerosis (MS), and more than 90% of children with spinal bifida suffer from NLUTD of varying severity. 4 In addition, NLUTD also arises in diabetes mellitus, unexpected sequelae after pelvic surgery, and cauda equina syndrome due to lumbar spine lesions. 4” in accordance to central nervous system disorder vs peripheral nervous system disorders. Page 4, line 77-80: I suggest to revise the sentences “NLUTD has a long course and the urinary system dysfunction progresses in a dynamic manner, which not only seriously affects patients’ QoL,5 but also easily leads to many complications that adversely affect the prognosis of patients, and even threaten the length of their life. 6” to: Neurogenic lower urinary tract dysfunction (NLUTD) not only threatens the health of affected patients long-term 6 but also has a significantly negative impact on the patients’ quality of life (QoL) 5. Page 4, line 80: Please use full percentages, i.e. 46 % instead of 45.59 %. Page 4, line 82: What “other upper urinary tract damages” are the authors referring to? Please specify or delete. Page 5, line 93: Remove (2018) Page 5, line 111: Remove (2019)

	Page 7, line 146: Why only adults? Any specific reason to exclude children and adolescences?

REVIEWER	Kantorová, Lucia The Czech National Centre for Evidence-Based Healthcare and Knowledge Translation, Institute of Biostatistics and Analyses
REVIEW RETURNED	21-Jul-2022

GENERAL COMMENTS	this is a very well drafted and designed protocol and the review has the potential to be exactly what you write. I have no major comments and only a few small suggestions below: line 142 Aim - It would be more accurate to use "this review" throughout the protocol. lines 65-68 - Please rephrase so it is clear you have decided to limit the search to English and Chinese languages. The statement currently is hard to understand, especially "we have no right to get access"... Line 185 - the inclusion of only recommendations "that describe the strength of evidence or the level of recommendation" requires some more explanation. Firstly, it would make more sense to include only recommendations that show BOTH the strength of evidence and the level of recommendation. Secondly, although guideline developers use various terms, the most correct would be using "certainty in the evidence" (or quality) and the "strength of recommendation". Nevertheless, please include some rationale for choosing to extract only such recommendations. Also, why not exclude guidelines that do not use systematic methods to depict the certainty in the evidence and the strength of recommendation, why do this at the level of recommendations? (And why do it at all?) You might consider separating recommendations that are based on scientific evidence from those that are based on expert evidence or consensus.

VERSION 1 – AUTHOR RESPONSE

Specific replies to Reviewer #1:Dr. Jose Medina-Polo, Univ Hosp 12 Octubre

Comments to the Author:

None

Answer：Thank you for taking the time to read this study.

Specific replies to Reviewer #2:Dr. Matthias Walter, Spinal Cord Injury Center & Research, University of Zurich

Comments to the Author:

The authors present a protocol to conduct a systematic review on clinical practice guidelines for the neurogenic lower urinary tract dysfunction (NLUTD). Their aim is clear, their methodology is sound, and their strengths and limitations are well balanced. Given the current paucity on that topic, their future systematic review will hopefully provide well-needed evidence to improve patient care. I have only minor comments.

1.Page 2, lines 23-25:

I suggest to revise the sentences to:

Neurogenic lower urinary tract dysfunction (NLUTD) not only threatens the health of affected patients long-term but also has a significantly negative impact on the patients' quality of life (QoL).

Answer：Thank you for your suggestion. It has been revised according to your suggestion. See lines 23-24 of the revised manuscript.

2.Page 2, line 26:

The authors use 'extant' several times throughout the manuscript. This reviewer feels that the author might not use the right word to express their intentions. Thus, the authors should either please explain what they are trying to indicate by using "extant" or use a different term instead.

- Page 3, line 54
- Page 5, line 106
- Page 6, line 133
- Page 6, line 137

Answer：Thank you for your suggestion. We used 'extant' several times in the first version of the manuscript to convey the meaning of 'current', and after consulting with a native English scholar, the word 'current' or 'present' would be more appropriate. We have replaced 'extant' with 'current' in the revised manuscript, which are as follows.

-line 26 'extant' is replaced with 'these'

-line 43 'extant' is replaced with 'current'

-line 129 'extant' is replaced with 'present'

-line 134 'extant' is replaced with 'current'

-line 357 'extant' is replaced with 'current'

3.Page 3, line 44:

With respect to "The findings of this study will be published in peer-reviewed journals", this reviewer feels that the authors meant to use journal, i.e. published in a peer-reviewed journal, rather than in multiple journals.

Answer：Thank you very much for your careful review of the text. According to your suggestion we have modified "published in a peer-reviewed journal" instead of "published in peer-reviewed journals", see line 47 in the revised manuscript.

4.Page 3, line 46:

With respect to the PROSPERO registration, there is a significant difference between the authors highlighted on PROSPERO compared to the current protocol submission. This should be clarified by the authors.

Answer：Thank you for your careful review. The reason for the discrepancy between the PROSPERO registration and the current protocol submission is that when we submitted the PROSPERO registration, our initial plan was that CZ, XW, JL, YM, SL, and WC would complete this systematic review, but JL, YM, and SL were unable to participate due to their scheduling factors. In fact, the idea for this systematic review and this manuscript was developed by CZ, YH, XW, FH, YW, LC and WC. In particular, (LC) and (WC) devised the study. (CZ) and (YH) drafted the protocol. (CZ, XW and FH) designed inclusion/exclusion criteria and search resources and strategies. (YH) and (YW) designed the evaluation strategy for each included guide. All authors provided critical feedbacks, helped shape the research and draft the manuscript.

5.Page 3, line 51:

With respect to “focused on a single population with NLUTD after spinal cord injury”, rather use focused on a single population with NLUTD, i.e. after spinal cord injury.

Answer：Thank you very much for your careful review of the text. According to your suggestion，we have modified “focused on a single population with NLUTD, i.e. after spinal cord injury”instead of "focused on a single population with NLUTD after spinal cord injury". However, as the editor considered that our presentation in the Strengths and Limitations section was too long, we have reworked it appropriately in this revised manuscript, see lines 52-64 in the revised manuscript.

6.Page 4, lines 70-72:

I suggest to revise the sentences, since “neurogenic bladder” is outdated and should not be used anymore, to:

Neurogenic Lower Urinary Tract Dysfunction (NLUTD) is defined by the International Continence Society (ICS) as a “lower urinary tract dysfunction due to disturbance of the neurological control mechanism”.

Answer : Thank you for your advice. According to your suggestion, we have revised it further. See lines 66-68 in the revised manuscript. In addition, “neurogenic bladder” is replaced by “ NLUTD” .

7.Page 4, lines 73-77:

Please revise this section “Studies show that 70%-84% of patients with spinal cord injury (SCI),2 53% of stroke patients,3 38%-71% of patients with Parkinson's disease, 32%-96% of patients with multiple sclerosis (MS), and more than 90% of children with spinal bifida suffer from NLUTD of varying severity. 4 In addition, NLUTD also arises in diabetes mellitus, unexpected sequelae after pelvic surgery, and cauda equina syndrome due to lumbar spine lesions. 4” in accordance to central nervous system disorder vs peripheral nervous system disorders.

Answer : Thank you for your valuable suggestions, we have modified them in accordance to central nervous system disorder vs peripheral nervous system disorders, as detailed in lines 68-74 of the revised manuscript.

8.Page 4, line 77-80:

I suggest to revise the sentences “NLUTD has a long course and the urinary system dysfunction progresses in a dynamic manner, which not only seriously affects patients’ QoL,5 but also easily leads to many complications that adversely affect the prognosis of patients, and even threaten the length of their life. 6” to:

Neurogenic lower urinary tract dysfunction (NLUTD) not only threatens the health of affected patients long-term 6 but also has a significantly negative impact on the patients’ quality of life (QoL) 5.

Answer : Thank you for your valuable suggestions. According to your suggestion we have modified. See lines 74-76 of the revised manuscript.

9.Page 4, line 80:

Please use full percentages, i.e. 46 % instead of 45.59 %.

Answer :Thanks to your suggestion. According to your suggestion we have modified. See line 76 of the revised manuscript.

10.Page 4, line 82:

What “other upper urinary tract damages” are the authors referring to? Please specify or delete.

Answer : Thank you for your careful review of the text, we have deleted the phrase "and other upper urinary tract damages" in the revised manuscript. See line 78 of the revised manuscript.

11.Page 5, line 93:

Remove (2018)

Answer : Thank you for your suggestion, we have removed (2018) in the revised manuscript. See line 90 of the revised manuscript.

12.Page 5, line 111:

Remove (2019)

Answer : Thank you for your suggestion, we have removed (2019) in the revised manuscript. See line 109 of the revised manuscript.

13.Page 7, line 146:

Why only adults? Any specific reason to exclude children and adolescences?
Answer : This is a very good question, which we have not considered well enough to present in the manuscript. Our reasons for including only adults consist mainly of the following:
Firstly, the etiology of NLUTD in children, adolescents and adults is very different, with congenital genetic disorders such as myelomeningoeele being more common in children and adolescents, while adult patients are usually caused by trauma and tumour; secondly, due to children's specific individual factors, many of the treatments available to adults are often limited in their application in children, so the treatment and management of children with NLUTD varies considerably. For example, clean intermittent catheterization is widely available in adult patients, but in children there is still controversy about when it should be performed and under what circumstances it should be initiated; bladder instillation drugs have been used in adults, but no data are available for children; Botulinum toxin injection have been used to treat adults with NLUTD, but have not yet been approved in children. Finally, national guideline development agencies also develop their guidelines separately according to the characteristics and differences of each population.
Considering these points, this review protocol excludes children or adolescent and only includes adult patients.

Specific replies to Reviewer #3:Dr. Lucia Kantorová, The Czech National Centre for Evidence-Based Healthcare and Knowledge Translation

Comments to the Author:
this is a very well drafted and designed protocol and the review has the potential to be exactly what you write. I have no major comments and only a few small suggestions below:

1.line 142 Aim - It would be more accurate to use "this review" throughout the protocol.
Answer : Thank you for your careful review of the text. In accordance with your suggestions, we have modified it, See line 137 of the revised draft. In addition, we have also carefully checked and revised other parts of the revised manuscript.

2.lines 65-68 - Please rephrase so it is clear you have decided to limit the search to English and Chinese languages. The statement currently is hard to understand, especially "we have no right to get access"...
Answer : Thank you for your careful review. In the revised manuscript, we have amended it to state "One of the limitations of this review is that only the CPGs that were published in English and Chinese will be included." See lines 60-61 and 366 of the revised manuscript for details.

3. Line 185 - the inclusion of only recommendations "that describe the strength of evidence or the level of recommendation" requires some more explanation. Firstly, it would make more sense to include only recommendations that show BOTH the strength of evidence and the level of recommendation. Secondly, although guideline developers use various terms, the most correct would be using "certainty

in the evidence" (or quality) and the "strength of recommendation". Nevertheless, please include some rationale for choosing to extract only such recommendations. Also, why not exclude guidelines that do not use systematic methods to depict the certainty in the evidence and the strength of recommendation, why do this at the level of recommendations? (And why do it at all?) You might consider separating recommendations that are based on scientific evidence from those that are based on expert evidence or consensus.

Answer : Thank you for your valuable suggestions.

Firstly, we do intend to include recommendations that include the quality of evidence and the strength of recommendation, the correct expression should be "that describe the quality of evidence and the strength of recommendation", which we have modified it, as detailed in lines 182 of the revised manuscript.

Secondly, we have replaced "the strength of evidence" and "the level of recommendation" in the text with "the quality of evidence" and "the strength of recommendation" respectively, as detailed in lines 39, 41, 182, 310 and 335 of the revised manuscript.

Thirdly, the quality of evidence can correctly measure the authenticity and credibility of the study, while the strength of recommendation can truly reflect the strength of recommendation of the clinical intervention. Therefore, by extracting recommendations that include both quality of evidence and strength of recommendation, we aim to present a concise and efficient overview of recommendations in different areas of NLUTD in order to provide decision-making support and guidance for evidence-based medicine and evidence-based practice.

Fourthly, we have added an exclusion criterion, which is "Guidelines that do not use systematic methods to depict the certainty in the evidence and the strength of recommendation", as detailed in lines 189-190 of the revised manuscript.

Fifthly, thank you very much for the advice you have given us. We will consider this carefully in our subsequent implementation of this systematic review. Our current plan is to extract recommendations that include both quality of evidence and strength of recommendation, but as recommendations in expert opinions and expert consensus do not include both quality of evidence and strength of recommendation, they have been excluded from this review protocol for the time being.

VERSION 2 – REVIEW

REVIEWER	Walter, Matthias Spinal Cord Injury Center & Research, University of Zurich, Neuro-Urology
REVIEW RETURNED	13-Oct-2022

GENERAL COMMENTS	All comments raised by this reviewer were addressed sufficiently by the authors.